🐍 PYTHON INTERVIEW QUESTIONS (Basic → Advanced)

🔹 SECTION 1: Python Fundamentals (Q1–Q10)

Q1. What is Python? Why is it called an interpreted language?

Answer:
Python is a high-level, general-purpose programming language.

Explanation:
Python is called **interpreted** because:

- Code is executed **line by line**
- No separate compilation step like C/C++

This makes debugging easier but execution slightly slower than compiled languages.

Q2. What are Python keywords? Can you name a few?

Answer:
Keywords are **reserved words** with special meaning.

Examples:
if, else, for, while, break, continue, return, class, def, try, except

Explanation:
You **cannot** use keywords as variable names because Python already assigns meaning to them.

Q3. How is Python dynamically typed?

Answer:
You don't need to declare variable types explicitly.

```
x = 10

x = "hello"
```

Explanation:
The variable x changes type at runtime.
Python decides the type **during execution**, not before.

Q4. What are mutable and immutable data types?

Answer:

Mutable	Immutable
list	int
dict	float
set	tuple
bytearray	string

Explanation:

- **Mutable** → Can change after creation
- **Immutable** → Cannot change

```
a = "hi"
a[0] = "H"  # ❌ error
```

Q5. What is the difference between = and ==?

Answer:

- = → Assignment
- == → Comparison

```
x = 5     # assignment
x == 5    # comparison
```

Explanation:
= stores a value
== checks equality

Q6. What are Python comments?

Answer:
Comments are ignored by the interpreter.

```
# Single line comment

"""
Multi-line comment
"""
```

Explanation:
Used to improve readability and explain logic.

Q7. What is indentation in Python and why is it important?

Answer:
Indentation defines code blocks.

```
if x > 0:
    print("Positive")
```

Explanation:
Python **does not use braces {}**.
Wrong indentation = IndentationError.

Q8. What is None in Python?

Answer:
None represents the **absence of a value**.

```
x = None
```

Explanation:
It is not 0, not False, not empty — it means **nothing assigned**.

Q9. What is type casting?

Answer:
Converting one data type to another.

```
int("10")

float(5)

str(100)
```

Explanation:
Used when data types don't match required operations.

Q10. What is the difference between print() and return?

Answer:

print	return
Displays output	Sends value back
No value stored	Value can be reused

```
def f():
  return 10
```

🔹 SECTION 2: Data Structures (Q11–Q20)

Q11. What is a list?

Answer:
An ordered, mutable collection.

```
nums = [1, 2, 3]
```

Explanation:

- Allows duplicates
- Index-based access

Q12. Difference between list and tuple?

List	Tuple
Mutable	Immutable
Slower	Faster
Uses []	Uses ()

Q13. What is a dictionary?

Answer:
Key-value pair data structure.

data = {"name": "Alex", "age": 30}

Explanation:
Keys must be **unique and immutable**.

Q14. What is a set?

Answer:
Unordered collection of unique elements.

s = {1, 2, 3}

Explanation:
Automatically removes duplicates.

Q15. How do you access dictionary values?

data["name"]

data.get("name")

Explanation:
get() avoids errors if key doesn't exist.

Q16. Difference between append() and extend()?

```
a.append([4,5])   # adds as one element
a.extend([4,5])   # adds individually
```

Q17. What is slicing?

```
arr = [1,2,3,4]
arr[1:3]
```

Explanation:
Extracts a portion of sequence.

Q18. What happens if key doesn't exist in dict?

```
data["x"]   # KeyError
```

Solution:
Use get().

Q19. How do you remove duplicates from a list?

```
list(set(my_list))
```

Q20. Difference between shallow copy and deep copy?

- Shallow → copies reference
- Deep → copies entire object

```
import copy
copy.deepcopy(obj)
```

🔹 SECTION 3: Control Flow & Functions (Q21–Q30)

Q21. What is a function?

Answer:
Reusable block of code.

```
def add(a, b):
  return a + b
```

Q22. What are positional and keyword arguments?

```
add(2,3)       # positional
add(a=2,b=3)     # keyword
```

Q23. What is *args?

Answer:
Handles variable number of positional arguments.

```
def f(*args):
  print(args)
```

Q24. What is **kwargs?

Answer:
Handles variable keyword arguments.

```
def f(**kwargs):
  print(kwargs)
```

Q25. What is recursion?

Answer:
Function calling itself.

```
def fact(n):
  return 1 if n==0 else n*fact(n-1)
```

Q26. What is a lambda function?

lambda x: x*x

Explanation:
Anonymous, single-expression function.

Q27. Difference between break and continue?

- break → exits loop
- continue → skips iteration

Q28. What is a pass statement?

Answer:
Placeholder that does nothing.

```
if x > 0:
  pass
```

Q29. What is enumerate()?

```
for i, v in enumerate(['a','b']):
  print(i, v)
```

Q30. What is zip()?

zip([1,2],[3,4])

Pairs elements together.

🔹 SECTION 4: Advanced Basics (Q31–Q45)

Q31. What is an iterator?

Answer:
Object that can be iterated using next().

Q32. Difference between iterator and iterable?

- Iterable → can be looped (list)
- Iterator → produces values (iter())

Q33. What is a generator?

```
def gen():
  yield 1
```

Memory-efficient iteration.

Q34. What does yield do?

Answer:
Pauses function and returns value without losing state.

Q35. What is exception handling?

```
try:
  x = 1/0
except ZeroDivisionError:
  print("Error")
```

Q36. Difference between is and ==?

- == → value equality
- is → memory identity

Q37. What is list comprehension?

[x*x for x in range(5)]

Q38. What is global vs local variable?

- Local → inside function
- Global → outside function

Q39. What is __name__ == "__main__"?

Ensures code runs only when file is executed directly.

Q40. What is Python memory management?

Handled by:

- Reference counting
- Garbage collection

Q41. What is a module?

Python file containing code.

Q42. What is a package?

Collection of modules.

Q43. What are docstrings?

"""Function description"""

Q44. What is input() used for?

Takes user input as string.

Q45. Why Python is popular in finance & data roles?

- Easy syntax
- Rapid prototyping
- Huge ecosystem